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Living with a Red Dwarf: X-Ray, UV, and Ca II Activity–Age Relationships of M Dwarfs

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Abstract

The vast majority of stars in the nearby stellar neighborhood are M dwarfs. Their low masses and luminosities result in slow rates of nuclear evolution and minimal changes to the stars' observable properties, even along astronomical timescales. However, they possess relatively powerful magnetic dynamos and resulting X-ray to UV (X–UV) activity, compared to their bolometric luminosities. This magnetic activity does undergo an observable decline over time, potentially making it a key age determinant for M dwarfs. Observing this activity is important for studying the outer atmospheres of these stars, but also for comparing the behaviors of different spectral type subsets of M dwarfs; e.g., those with partially versus fully convective interiors. Beyond stellar astrophysics, understanding the X–UV activity of M dwarfs over time is a key component when studying the atmospheres and habitability of any hosted exoplanets. Earth-sized exoplanets, in particular, are more commonly found orbiting M dwarfs than any other stellar type, and thermal escape (driven by the M dwarf X–UV activity) is believed to be the dominant atmospheric loss mechanism for these planets. Utilizing recently calibrated M dwarf age–rotation relationships, also constructed as part of the *Living with a Red Dwarf* program, we have analyzed the evolution of M dwarf activity over time, in terms of coronal (X-ray), chromospheric (Lyα, and Ca II), and overall X–UV (5–1700 Å) emissions. The activity–age relationships presented here will be useful for studying exoplanet habitability and atmospheric loss, and also for studying the different dynamo and outer atmospheric heating mechanisms at work in M dwarfs.

Unified Astronomy Thesaurus concepts: Stellar ages (1581); Stellar atmospheres (1584); Stellar coronae (305); Stellar chromospheres (230); Stellar evolution (1599); Stellar rotation (1629); X-ray astronomy (1810); Ultraviolet astronomy (1736); Low mass stars (2050); Late-type dwarf stars (906); M dwarf stars (982)

Supporting material: machine-readable tables

1. Introduction and Background: Studying M Dwarfs

Magnetic fields are theorized to exist around all cool, main-sequence stars, as massive as late F dwarfs down through late M dwarfs (collectively referred to here as FGKM dwarfs). These fields are responsible for (or contribute to) a range of observable behaviors (e.g., heating the outer stellar atmosphere to 10^5–10^6 K). These heated plasmas comprise the (if structured similar to the Sun) stellar chromospheres, transition regions, and coronae. The mechanism responsible for generating these magnetic fields is the dynamo effect, which involves contributions from both convective motions and stellar rotation, though in varying amounts depending on the mass of the star (described later). The combination of FGKM dwarf magnetic fields and stellar winds are also responsible for the spin-down effect. First discovered just over 50 yr ago (Skumanich 1972), this effect results in the lengthening of an FGKM dwarf's rotation period and the weakening of its dynamo as it ages.

Thus, the rotation periods and/or magnetic activity levels of FGKM dwarfs can serve as age determinants. The age–rotation relationships we have constructed for M0–6.5 dwarfs as part of the *Living with a Red Dwarf* program have been described in a companion paper (Engle & Guinan 2023). This paper focuses on M dwarf activity–age relationships.

Several observable proxies exist for measuring a star's level of magnetic activity. Coronal emissions are regularly characterized by observations at soft X-ray energies. The soft X-ray band encompasses 0.1–10 keV although, given the sensitivity of most current X-ray instruments and the narrow energy range over which quiescent FGKM dwarf coronae emit, most observed fluxes are measured in a somewhat narrower energy range (e.g., 0.2–3 keV or 0.2–5 keV). Emissions from the transition region or chromosphere are often characterized by a range of spectral features at UV and optical wavelengths. Even though these optical spectral features exist that probe plasmas with chromospheric temperatures, the majority of chromospheric emission features are observed in the UV range. For this reason, we will use the term X–UV primarily to refer to the outer atmospheric (coronal–chromospheric) emissions or activity of M dwarfs. The astrophysical importance of measuring these quantities has prompted widespread use in many studies. Though this is a very active field with far too many studies to allow a comprehensive list, a selection of recent works involving rotation rates and/or atmospheric activity levels include Wright et al. (2011), Stelzer et al. (2013), Barnes et al. (2016), Guinan et al. (2016), Youngblood et al. (2017), Suárez Mascareño et al. (2018), Curtis et al. (2020), Linsky et al. (2020), Magaudda et al. (2020), Johnstone et al. (2021), Youngblood et al. (2021), Ibañez Bustos et al. (2023), and Marvin et al. (2023).

For this study, we have constructed X–UV activity–age relationships for M0–M6.5 dwarfs (M dwarfs hereafter), using ages determined with the age–rotation relationships presented in Engle & Guinan (2023). These relationships are not only valuable for the information they provide about the stars

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themselves, but they also delineate the space environments that M dwarfs create for any exoplanets orbiting them.

This information is invaluable for multiple reasons. The first is that M dwarfs represent ∼75% of the nearby stellar inventory (Reylé et al. 2021)—studying M dwarfs allows us to understand the behavior of the Universe's largest stellar population. Also, M dwarfs are less massive ($M \approx 0.6$–$0.1\ M_{\odot}$), smaller ($R \approx 0.6$–$0.1\ R_{\odot}$), and cooler ($T_{\rm eff} \approx 3900$–2850 K) than the Sun, and have much lower luminosities ($L \approx 0.06$–$0.001\ L_{\odot}$).[1] Their low masses result in slow core reaction rates and long, seemingly stable lifetimes (∼100 Gyr up to as long as $\sim 10^{12}$ yr; Choi et al. 2016). While their bolometric luminosities are low, and experience little change over billions of years, the X–UV activity of M dwarfs is comparatively strong and highly variable. This can severely impact hosted exoplanets that would otherwise need to orbit near to the (low-luminosity) star to maintain a temperate surface. As stellar X–UV emissions are substantial drivers of both photochemical reactions within the planetary atmospheres and also the loss of the planetary atmospheres, determining the evolution of these emissions with age has had a broader impact on the field, increasing the importance of such studies.

The paper is organized as follows. In Section 2, we briefly discuss the issues surrounding the potential habitability of planets orbiting M dwarfs. In Section 3, we present and describe the data used for this study. In Section 4, the activity–age relationships will be provided and discussed. Finally, in Section 5, we will provide a brief summary of our results.

2. Orbiting M Dwarfs: Can They Host Habitable Planets?

Whether M dwarfs can host habitable planets is a question that has received considerable attention from the research community. For a thorough, recent review of the numerous factors influencing the habitability of planets orbiting G, K, and M dwarfs see Airapetian et al. (2020) and references therein. Although not the focus of this paper, since X–UV activity plays a prominent role in exoplanet atmospheric evolution/loss/retention and habitability studies, we will provide a (very) short summary of the topic. Some of the major aspects governing potential M dwarf exoplanet habitability can be classified as positives, negatives, and unknowns.

"The positives" mainly involve simple statistics. M dwarfs represent the most numerous stellar component of observed space, live extremely long main-sequence lifetimes with near-negligible changes in either size, optical luminosity, or surface temperature, and studies have thus far shown them to host terrestrial-size planets at a higher rate than more massive stars. All considered, even if the odds are overall very slim that M dwarfs can host habitable planets, there are so many to consider that perhaps some exoplanets have still managed to evolve with the necessary combination of characteristics that allow them to support habitable environments. It is also worth noting that M dwarf exoplanets are ideal targets for atmospheric composition studies via transmission spectroscopy due to their relatively deep transits when compared to other more massive (luminous) spectral types. Though this last point is not a "positive" for the actual habitability of the planet, it is a benefit when studying the planet's potential habitability and has also generated further interest in the field.

"The negatives" center on the planetary consequences of M dwarf magnetic activity. Due to their lower bolometric luminosities, their HZs are very close (∼0.07–0.4 au for M5—M0 V stars) so that tidal locking and possible tidal heating may take place (Jackson et al. 2008). With the planet orbiting so near to its host star, it is exposed to more intense X–UV emissions, along with flares and coronal mass ejections (CMEs). The end result can be the stripping (erosion) of the planet's atmosphere and sterilization of its surface.

"The unknowns" primarily involve finer resolution of M dwarf activity, and at present almost all details about the planets themselves. The more intense X–UV radiation that potentialy habitable exoplanets around M dwarfs will receive has already been mentioned as an overall negative. However, there is the additional unknown component of the exact role that near-ultraviolet (NUV) radiation played in prebiotic chemistry on Earth (Ranjan et al. 2017). It is possible that M dwarfs are not luminous enough in the NUV to drive such chemistry, although their high flare frequencies could help compensate for this (Froning et al. 2019; Brown et al. 2023). All considered, much more research is still needed on these topics. The reader is again directed to Airapetian et al. (2020) for a more thorough discussion, but there are numerous paths along which both the formation and subsequent evolution of the planet and its atmosphere can proceed. M dwarfs have higher luminosities as they are still collapsing and evolving onto the main sequence. Some studies show this could force HZ planets into a runaway greenhouse environment, although (depending on the conditions within the nebula from which the star–planet system formed) the planets could be protected from this. The strength/extent of the planet's magnetic field and whether it can offer adequate protection is another important unknown (Ribas et al. 2016), as is the amount of atmospheric replenishing that occurs due to volcanic outgassing. Related to this last unknown are the results of France et al. (2020), proposing that planets around older and less active M dwarfs (their study focused on GJ 699, aka Barnard's Star, age ≈9.5 Gyr, and a target within this study) could develop "second-generation" atmospheres via outgassing and at last become habitable worlds.

Particularly important unknowns on the stellar side are robust delineations of both M dwarf flare frequencies and CMEs as they evolve over time. The comprehensive study by Wood et al. (2021) finds that "CMEs from M dwarfs may be much less common than generally thought, despite the high flare rate, so perhaps CME exposure is not as big a factor for habitability as often supposed." Even the habitability of a tidally locked planet can remain in question due to unresolved planetary characteristics such as cloud cover, atmospheric circulation, and surface arrangement.

Although the field is still some time away from precisely determining the conditions present in a wide range of exoplanet atmospheres, surfaces, and interiors, we have at last developed a method for determining the ages of the host stars (and thus their coeval planetary systems) and can more precisely track the X–UV evolution of M dwarfs over time.

3. M Dwarf Ages and X–UV Activity Measures

3.1. A Brief Discussion of Target Ages

We will begin by quickly summarizing the sources of M dwarf ages used in this study. The age–rotation relationships

[1] https://www.pas.rochester.edu/~emamajek/EEM_dwarf_UBVIJHK_colors_Teff.txt

of Engle & Guinan (2023) were used to determine most of the stellar ages used in this study. For a limited number of targets a reliable, previously determined age was available and this age was used instead. These targets are either members of star clusters/associations, belong to the thick disk or halo populations of the Milky Way, or have a companion object for which an alternative age could be determined and applied to the M dwarf. The data initially displayed two evolutionary paths for M dwarf rotation periods (Guinan & Engle 2018), with the division occurring between spectral types M2 and M2.5. This division was confirmed as additional data were added to the relationships, and is likely caused by the M dwarfs on either side of this divide operating under different dynamo mechanisms. A similar spectral type division was also found in the activity measure data of Mullan & Houdebine (2020). For ages below ∼2.5–3 Gyr, a further subdivision between M2.5–3.5 and M4 to ∼M6.5 stars is apparent in the cluster rotation rate data, and this subdivision is most likely due to the much longer pre–main-sequence lifetimes of the M4–6.5 subset.

The age–rotation relationships of all M dwarf subsets show an inflection point, sharply dividing each relationship into a first (younger) and second (older) track. This is in agreement with previous studies (see Curtis et al. 2020), who referred to the end of the younger track as a stalling of the spin-down effect, before resuming again along the older track. A proposed theory for why this occurs is that zero-age main-sequence M dwarfs initially have a strong differential rotation profile within their interiors. Along the younger track, the stellar interior is synchronizing and transfer of angular momentum from the interior compensates for that which is lost from the surface (Spada & Lanzafame 2020). The younger track ends with the resynchronization of the stellar interior, when the star begins rotating as a solid body, and a new track of rotational evolution begins. For stars in this study whose rotation periods place them on the "older" tracks, ages and uncertainties from both the segmented and orthogonal distance regression (ODR) fits (see Figure 1 and Engle & Guinan 2023) were combined into a mean value and error of the mean.

3.1.1. A Note on Target Spectral Types

The targets have been divided into subsets based on spectral type. As Magaudda et al. (2020) presented the largest recent database of M dwarfs with activity measures, their stellar parameters (derived using the empirical relationships of Mann et al. 2015) are used for many of the targets in this study. However, the literature was also searched for spectral type measures of all stars from Magaudda et al. (2020) as a check, and to determine the spectral type for targets presented in this study that were not in Magaudda et al. (2020). Preference was usually given to studies that either were more recent (e.g., included updated model atmospheres), employed spectral or spectral energy distribution (SED) fitting, or specifically focused on the determination of M dwarf parameters instead of a wider range of (or all) stellar types. For all tables presented here, if (depending on the spectral type source that was used) a target could have potentially been placed in the other M dwarf subset, that target's name will be in bold.

3.2. X–UV Activity Measures

X-rays, and most wavelengths of UV radiation, are not able to reach the Earth's surface and thus require observations from space-based or high-altitude (balloon-borne) observatories. As these are very high-quality and well-calibrated instruments, the data they produce are second to none. Their observing time, however, is both limited and highly sought after (oversubscribed). Though oversubscription of these instruments is understood and unavoidable, it raises the difficulty in repeatedly measuring multiple targets over time which would greatly benefit the X–UV relationships. The main issue is that stellar coronal–chromospheric activity is variable over timescales of hours (flares), days to months (stellar rotation), years (magnetic activity cycles), and finally megayears to gigayears (weakening of the stellar magnetic dynamo due to spin-down). Altogether, the amplitudes of variation involved can be two orders of magnitude or more for the most extreme cases. Single observations of sufficient exposure time can allow the observer to avoid contamination by stellar flares, which can cause some of the highest-amplitude variations. The issue still remains that, until a target has been observed multiple times over a significant time span, the observer is left with an incomplete measure of the target's mean activity level and this effect will likely propagate into higher uncertainties within the activity–age relationships.

The X-ray database currently has two advantages. First, measurements do not require high spectral (energy) resolution, allowing observations to be gathered more efficiently. Second, the German–US–UK collaborative Röntgen Satellite (RÖSAT) was designed for efficient X-ray observations and carried out numerous observations of stars in the 1990s, including an invaluable all-sky survey. Observations with modern X-ray missions (e.g., Chandra, XMM-Newton, Neil Gehrels Swift Observatory, and very recently eROSITA) can be added to this previous data if available, allowing us to characterize the range of activity (and mean activity level) for each target better.

The majority of UV activity from cool dwarfs can be attributed to emission lines. Specifically for M dwarfs, a single feature (the Lyα 1216 Å [Lyα] line) is responsible for 50% of the total UV emissions or more (France et al. 2012). Due to its relative strength, this line has naturally become a high-priority target for observers, but a successful analysis requires data with medium to high spectral resolution and/or an instrument that can successfully mitigate geocoronal contamination. Interstellar absorption features of hydrogen and deuterium occur at the same wavelengths, and these need to be modeled and accounted for before the "pure" stellar Lyα flux can be obtained. The instrument and analysis requirements result in a more limited database of reliable Lyα measures, and most of those presented here were obtained from the literature and measured in the previously described fashion by the MUSCLES and mega-MUSCLES Hubble Space Telescope observing programs (see France et al. 2016; Loyd et al. 2016; Youngblood et al. 2016; Froning et al. 2019; Wilson et al. 2021).

Fortunately, there are also a number of cool dwarf magnetic activity "tracers" observable at optical wavelengths. As in the UV, this activity is observed through emission (sometimes absorption) lines. Though popular alternatives exist (e.g., Hα and Na I), we have selected the Ca II H and K features for this study. The behavior of these features in cool stars has been studied for over 40 yr (Vaughan & Preston 1980), and they occur at the short end of the optical spectrum (∼3930–3970 Å), where M dwarf photospheric contributions will be minimal. Even so, a reliable and well-studied quantity ($R'_{\rm HK}$) has been

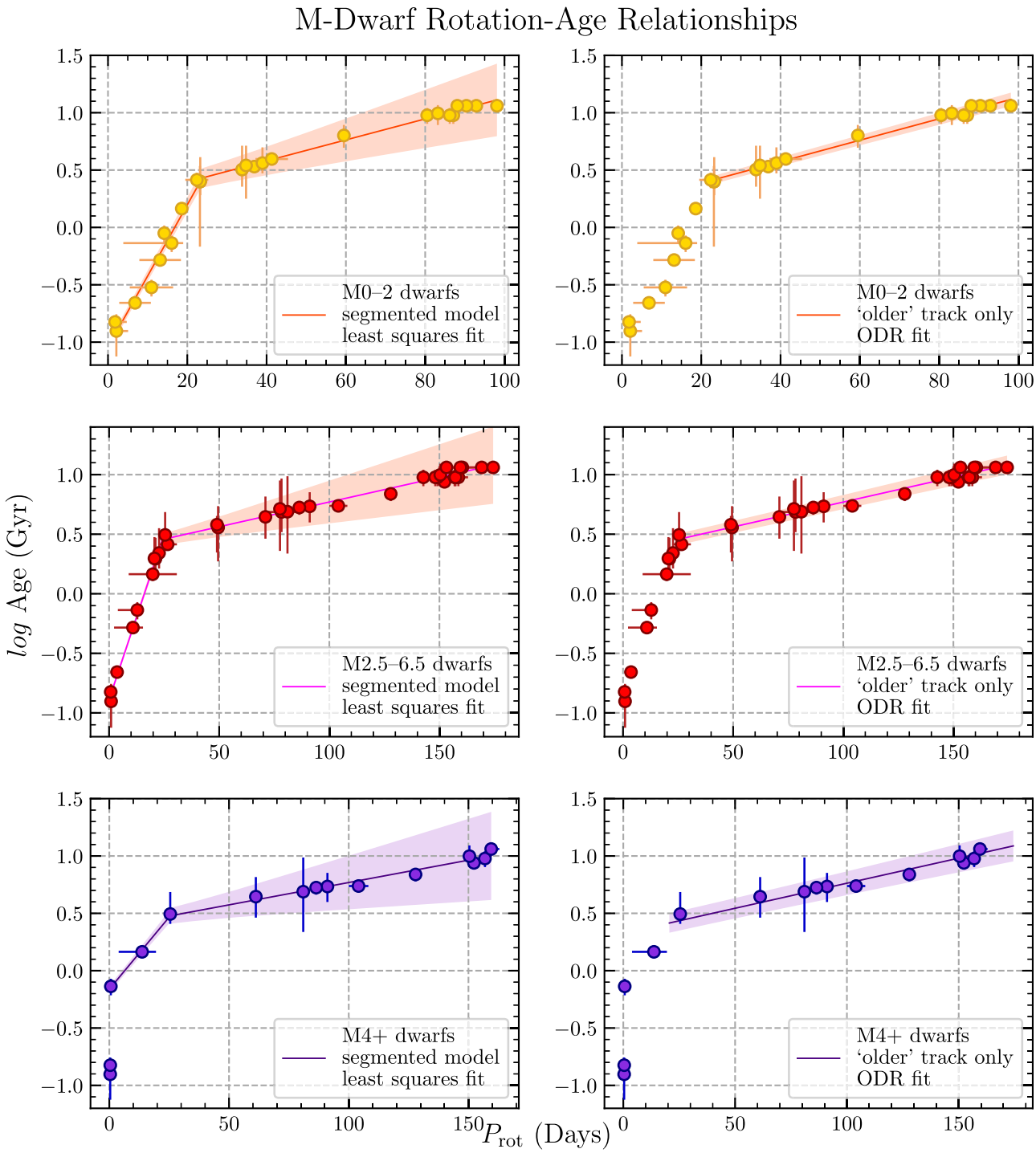

Figure 1. M dwarf age–rotation relationships from the *Living with a Red Dwarf* program (Engle & Guinan 2023). With the exception of cluster or galactic population members, these relationships were used to calculate the stellar ages used in this study and presented in the tables. Each row of plots is devoted to a specific subset of M dwarfs. The "early" M dwarfs, M0–2 dwarfs, are plotted in the top row. The "mid–late" M dwarfs are plotted in the middle row. For the "older" track, M2.5–6.5 dwarfs are plotted together since they have all settled onto a common evolutionary path. However, only M2.5–3.5 dwarfs are plotted on the "young" track due to the large differences in pre–main-sequence lifetimes that are encountered within this mass range. Consequently, a third subset was created and shown in the bottom row. Here, only M4 (and later) dwarfs are plotted. As seen, there is a large difference between the young tracks of the middle and bottom rows, but any difference between the older tracks is well within the uncertainties of the fits.

derived for these features which involves removing the photospheric contribution to allow an easier comparison of different spectral types. Still, a recent study by Marvin et al. (2023) found that offsets can occur depending on the exact methodology used over time. When applicable, the corrections of Marvin et al. (2023) were applied to target data before the representative value presented here was determined. After applying the correction, comparing the values determined in this study to those of Marvin et al. (2023) returned an average difference of $\Delta \log R'_{\rm HK} \approx 0.031$.

The clear advantage of optical magnetic activity tracers like Ca II H and K is that they can be observed using ground-based

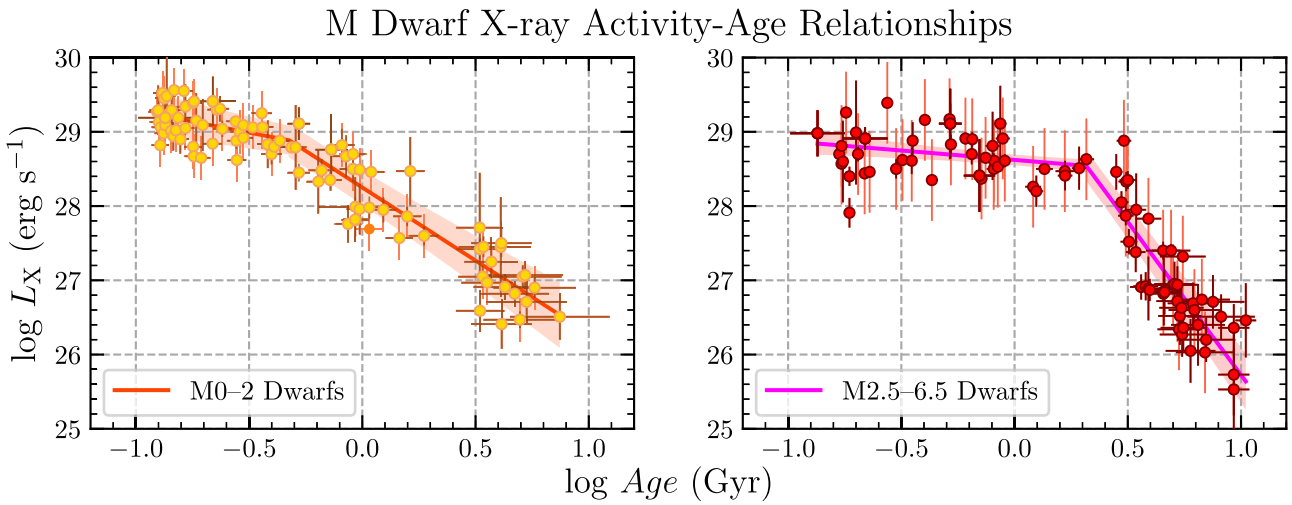

Figure 2. The evolution of M dwarf X-ray activity over time. Early M dwarfs are plotted on the left, and mid–late M dwarfs on the right. Lighter-colored error bars indicate when the "average" uncertainty value was applied to a target for fitting purposes. A two-component, segmented linear model was applied to each subset and is plotted. Each subset shows an initial "saturation phase" where high activity levels are sustained for a period before an inflection point is reached. After this, the activity decreases at an accelerated rate. It is worth noting that many studies assume a constant level of activity during the saturation phase, where our fits indicate that a slight decrease occurs. The saturation phase is found last ∼470 Myr for the early M dwarfs and ∼2.1 Gyr for the mid–late M dwarfs.

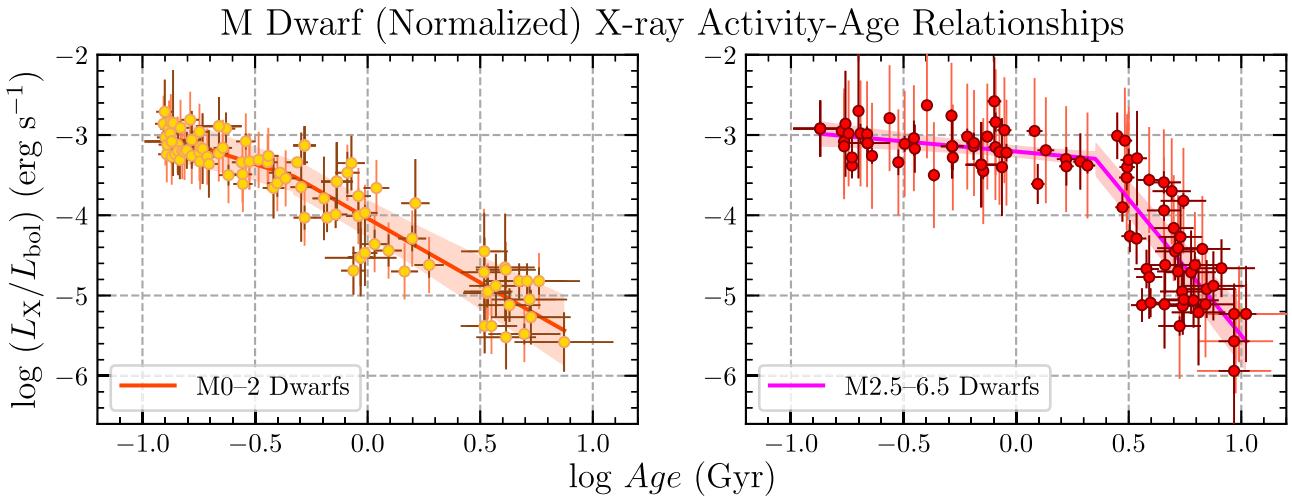

Figure 3. The evolution of M dwarf X-ray activity over time is again plotted, but now the ratio of X-ray to bolometric luminosity is used to normalize the stars better. Again, lighter-colored error bars indicate when the "average" uncertainty value was applied to a target for fitting purposes. As revealed by the segmented fits to the data, both subsets begin their saturation phases at essentially equal activity levels ($\log(L_X/L_{bol}) \approx -3$), and decline by ∼2.5 orders of magnitude. The length of the saturation phase is found to be ∼450 Myr for the early M dwarfs and ∼2.3 Gyr for the mid–late M dwarfs.

instruments with lower oversubscription rates. Still, medium- to high-resolution spectroscopy is required and the faintness of the targets can be prohibitive for smaller telescopes, especially when observing M dwarfs at short optical wavelengths. Previous spectroscopic surveys were dedicated to such activity measures, but recent planet-hunting surveys (e.g., with the HARPS instrument) have produced a large amount of high-quality spectra that have also been used for activity measures, many of which were included in this study (see Perdelwitz et al. 2021).

4. The M Dwarf Activity–Age Relationships

Here we present the relationships constructed by the *Living with a Red Dwarf* (*LivRed*) program, along with a discussion of the methods used to develop them and their results. As was done previously for the rotation–age relationships, a two-segment linear equation was defined via `numpy.piecewise`,

$$\text{Activity index} = \begin{cases} a \times \log \text{Age(Gyr)} + b & [\text{for} \log \text{Age} < d], \\ a \times \log \text{Age(Gyr)} + b + c \times (\log \text{Age(Gyr)} - d) & [\text{for} \log \text{Age} \geqslant d], \end{cases} \quad (1)$$

and then fit to each data set using `scipy.optimize.least_squares`. Within the sample of each activity index, an average "scatter" value was calculated using the targets with multiple measures, and this value was used as the uncertainty for single-measure targets when fitting the data. This application of a larger uncertainty value than most targets' data assumes that all targets have variable activity levels, but only those with multiple measures have given an insight into what the average amplitude of those variations should be. In Figures 2, 3, 4, and 5, the lighter error bars show instances

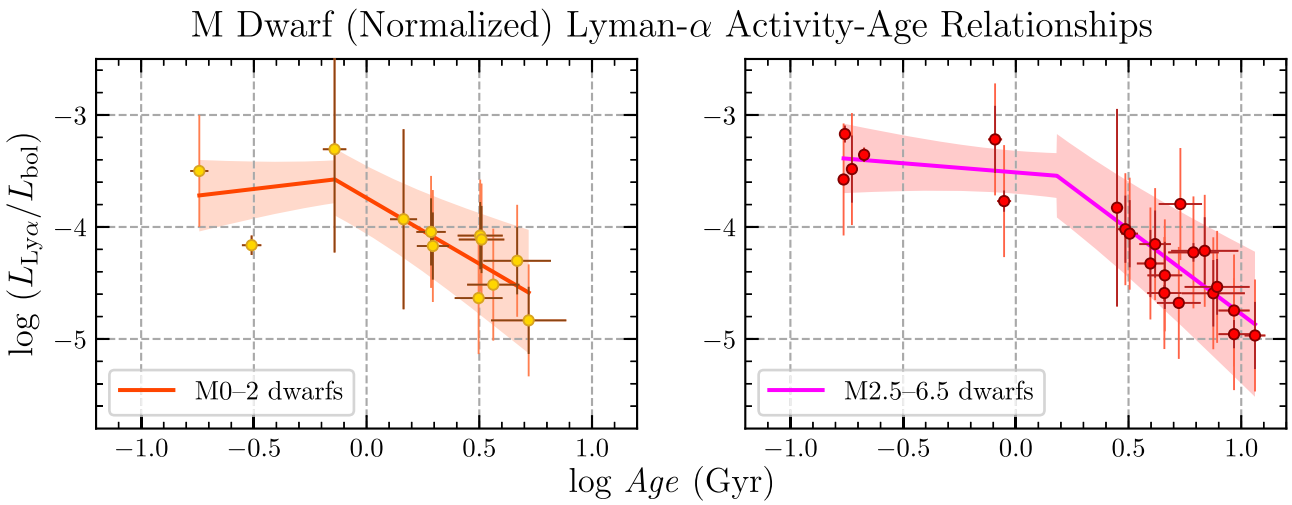

Figure 4. The evolution of Lyα activity over time is plotted. (Again, lighter-colored error bars indicate when the "average" uncertainty value was applied to a target for fitting purposes.) The reliability of the fits is impacted by the lack of data, particularly for the early M dwarfs. The current data show saturation phase lengths of ∼720 Myr for early M dwarfs and ∼1.5 Gyr for mid–late M dwarfs. Again, however, these quantities should not be regarded as highly reliable due to the lack of data. Instead, we would advise using the Ca II relationship (Figure 5) as a more reliable indicator of chromospheric activity over time.

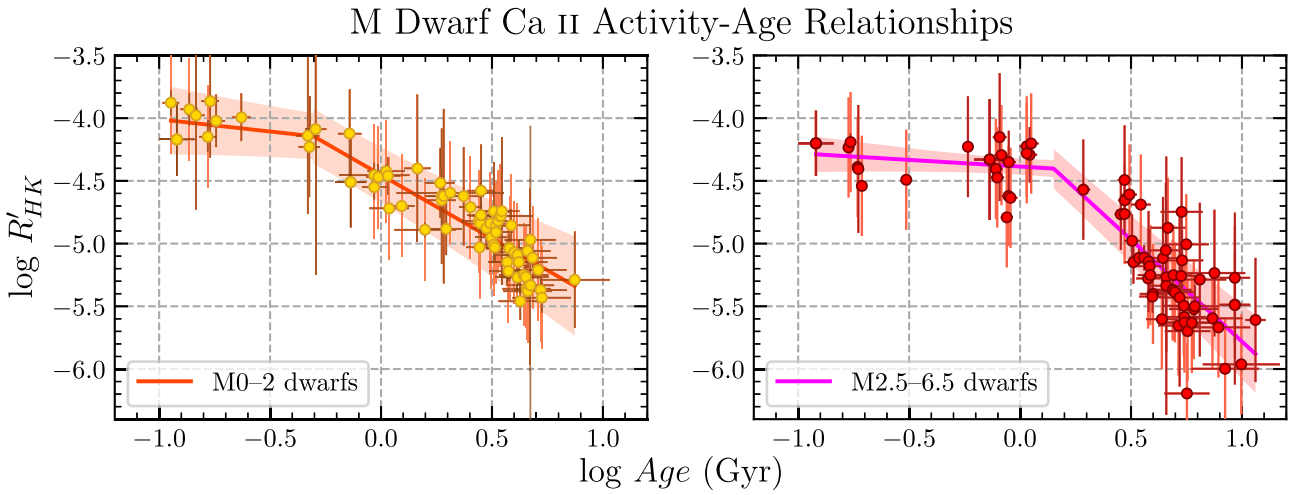

Figure 5. Evolution of Ca II H and K emission over time, as given by the $\log R'_{\rm HK}$ index. (Once again, lighter-colored error bars indicate when the "average" uncertainty value was applied to a target for fitting purposes.) The Ca II emission line cores form within the stellar chromosphere and can thus serve as a proxy for cooler UV features, such as Lyα. Given the richer data set, the Ca II relationships are also more reliable than those for Lyα. Both stellar subsets are observed to decline by ∼1.5 orders of magnitude, a notably less drastic decline than shown for X-ray activity, with saturation phase lengths of ∼500 Myr for early M dwarfs and ∼1.4 Gyr for mid–late M dwarfs. The early M dwarf saturation phase length can be considered comparable to the X-ray, given the uncertainties. The mid–late M dwarfs show a considerable difference in saturation phase lengths at present. It is important to note, however, that although the Ca II data set is far richer than Lyα, one age range that is not well covered occurs right near the inflection point. Further data at this vital age range will be necesary to pin down the chromospheric saturation phase length reliably in comparison to the coronal, as shown by the X-ray data. Another notable feature of these relationships is that early M dwarfs appear to exhibit higher relative levels of chromospheric activity at both young and old ages. However, much of this difference can be accounted for by the uncertainties and scatter of the fits and data.

where this average scatter value has been applied, and the darker error bars show the uncertainty based on actual data of the target itself.

Activity–age relationships have been constructed in the X-ray, UV, and for the short optical wavelength Ca II emission line doublet. The Ca II observational database is much richer than that of Lyα, and the two emission features form in plasmas of overlapping temperatures and stellar atmospheric regions (though the Lyα region extends to higher temperatures/altitudes, while the Ca II regions extends to lower temperatures/regions; Vernazza et al. 1981; Hall 2008). These relationships serve two purposes: they provide additional age determinants for M dwarfs, as we will discuss, and they also delineate the average high-energy (X–UV) fluxes to which exoplanets orbiting such stars have been, are presently being, or will in the future be subjected. Further, thoroughly studied relationships exist between Ca II, Lyα, and numerous other UV emission features (e.g., see Melbourne et al. 2020; Pineda et al. 2021, and references therein). These could allow any or all of the relationships (or combinations thereof) presented both here and in Engle & Guinan (2023) to be utilized in refining an M dwarf's age, and the high-energy environment to which an exoplanet orbiting it would be subjected.

The X-ray relationships are plotted in Figure 2 (X-ray luminosity over time) and Figure 3 (the ratio of X-ray to bolometric luminosity over time), and the best-fitting

Table 1
Best-fitting Activity–Age Parameters

	M0–2		M2.5–6.5	
	Parameter	Uncertainty	Parameter	Uncertainty
		$\log L_X$		
a	−0.5836	0.2447	−0.2534	0.1505
b	28.6993	0.1787	28.6213	0.0726
c	−1.3858	0.2689	−3.8713	0.3988
d	−0.3240	0.0902	0.3171	0.0487
		$\log L_X/L_{bol}$		
a	−0.8820	0.2444	−0.2542	0.1591
b	−3.8021	0.1784	−3.2064	0.0768
c	−0.7164	0.2685	−3.1416	0.4184
d	−0.3314	0.1729	0.3516	0.0611
		$\log L_{Ly\alpha}/L_{bol}$		
a	0.2081	1.7016	−0.1619	0.3197
b	−3.5663	0.3692	−3.5114	0.1914
c	−1.3478	0.9079	−1.3472	0.4673
d	−0.1420	0.3436	0.1809	0.2333
		$\log R'_{HK}$		
a	−0.1929	0.2205	−0.1062	0.1354
b	−4.2022	0.1713	−4.3860	0.0648
c	−0.8214	0.2329	−1.5184	0.2502
d	−0.3053	0.1487	0.1512	0.0942
		$\log(L_{X-UV}(5\text{–}1700\text{Å})/L_{bol})$		
a	−0.4896	0.1266	−0.1456	0.0911
b	−3.2128	0.0914	−2.8876	0.0439
c	−0.4469	0.1441	−1.8187	0.2412
d	−0.2985	0.1528	0.3545	0.0604

Note. All equations are of the format: activity index $= a \times \log \text{Age(Gyr)} + b$ [for$\log\text{Age} < d$] and activity index $= a \times \log\text{Age(Gyr)} + b + c\times(\log \text{Age(Gyr)} - d)$ [for $\log\text{Age} \geqslant d$].

parameters are presented in Table 1. The data used to construct these relationships are provided in Tables 2 and 3. We have endeavored to gather multiple measures of X-ray activity for each star whenever possible, to help mitigate scatter due to stellar variability, though this does not eliminate the issue as many of the stars still only possess one or two measures. As further measures of the targets are carried out by current and future X-ray missions, it will be interesting to see how (or if) the remaining scatter in Figures 2 and 3 is reduced. M dwarf ages were calculated using the age–rotation relationships presented in Engle & Guinan (2023), and discussed in Section 3.1, unless there was an existing age determination (e.g., membership in a cluster or the thick disk/halo populations). As seen in the plots, the sample is separated into the same "early" and "mid–late" subsets used for the rotation relationships, and the X-ray activity of both M dwarf subsets decreases by ∼2.5–3 orders of magnitude over the plotted age range (∼0.1–12.5 Gyr).

The most notable difference between the M dwarf subsets is the substantially shorter "saturation phase" of early M dwarfs, compared to the mid–late subset. Though many previous studies fixed the saturation phase to a flat level of activity, we instead opted to leave all parameters as free. In both subsets, there appears to be a slight decline in activity over the duration of the saturation phase, though it is within the scatter of the data. As with the age–rotation relationships, we advise restraint when using the first "track" of the X-ray relationships to quote a precise age. Here, however, is where one aspect of the early subset relationship's usefulness is found. After ∼0.5 Gyr, early M dwarfs exit their saturation phase and X-ray activity levels begin to decline more rapidly. This establishes a range of ages where, although the rotations of early M dwarfs are still evolving along the less reliable, first (resynchronizing) track, their X-ray activity has shifted onto its second track, where more reliable ages can be determined. In effect, the X-ray relationship extends the range over which more reliable ages can be determined for early M dwarfs. Additional methods of age determination involving kinematics (Lu et al. 2021) and abundances (Carrillo et al. 2023) can also be employed to achieve results based on as many techniques as the data allow. This opportunity to determine M dwarf ages through multiple methods, or to compare the ages of each method for targets with high-quality data, holds great potential for future studies.

Although the mid–late subset appears to begin its saturation phase at a slightly lower X-ray luminosity than the early subset (∼28.9 versus 29.2 erg s^{-1} cm^{-2}, according to the fits (Figure 2), though within the scatter), it is important to note that both subsets show essentially the same initial levels of X-ray activity when normalized according to bolometric luminosity (see Figure 3). Also, the mid–late M dwarfs remain in their saturation phase for as long as ∼2.3 Gyr. This presents a far more extended period over which mid–late M dwarfs sustain enhanced levels of X-ray activity, making the ages determined via this method less reliable over a larger age range, and any exoplanet orbiting a mid–late M dwarf must endure an initial phase of high X-ray activity that is ∼4–5× longer than it would be if the planet were orbiting an early M dwarf. The consequences of this phase on the planet's atmosphere, even potentially affecting its ability to retain elements heavier than H/He (see Ribas et al. 2016; Shields et al. 2016), would be particularly relevant to M dwarfs and stands to benefit from additional theoretical work.

This dramatic difference in saturation phase length is also interesting in terms of stellar astrophysics. Early M dwarf X-ray activity begins to decline shortly after the star arrives on the main sequence, but mid–late M dwarf X-ray activity remains saturated until they are nearly finished evolving along their first rotation track. This is likely a result of the different dynamo mechanisms at work within the two subsets, and the relative contributions of rotation and convection toward each one. As described by Mullan & Houdebine (2020), there are three stellar magnetic dynamos for which models exist—$\alpha\Omega$, α^2, and $\alpha^2\Omega$ (where α represents the contribution due to convective turbulence [in the presence of rotation, giving rise to Coriolis force effects], and Ω represents angular velocity). The more massive early M dwarfs are theorized to operate under the $\alpha\Omega$ dynamo—similar to the Sun—and this would explain why the spin-down effect, and decreasing angular velocity, would quickly result in weakening magnetic fields and lower X-ray activity levels. The mid–late M dwarfs, however, generate magnetic fields via either an $\alpha^2\Omega$ or α^2 dynamo. Due to the diminished contribution of angular velocity, mid–late M dwarfs can evolve though >2 Gyr of spin-down, and associated angular velocity loss, without any

Table 2
Early M Dwarf X-Ray Data

Star Name	P_{rot}	Uncert.	P_{rot} src	Log Age	Unc.	$\log(\langle L_X \rangle)$	Unc.	$\log(\langle L_X \rangle / L_{bol})$	Unc.	X-Ray src
1RXS J041330.7+251052	2.22	0.02	M20	−0.906	0.038	29.27	0.08	−2.86	0.10	M20
EPIC 201909533	2.29	0.02	M20	−0.901	0.038	29.29	0.33	−2.71	0.40	R, M22
1RXS J030708.3+323528	2.35	0.02	M20	−0.898	0.038	29.13	0.06	−3.03	0.07	M20
EPIC 210651981	2.44	0.02	M20	−0.892	0.038	28.82	0.05	−3.24	0.06	M20
1RXS J231233.3+391140	2.47	0.02	M20	−0.890	0.038	29.06	0.11	−3.12	0.13	M20
GSC 03517-00212	2.61	0.03	M20	−0.882	0.039	29.52	0.02	−2.92	0.02	M20
UCAC4 630-046962	2.69	0.03	M20	−0.877	0.039	28.98	0.03	−2.98	0.04	M20
CK Tri	2.73	0.03	M20	−0.874	0.039	29.45	0.01	−2.84	0.01	M20
LP 152-39	2.79	0.03	M20	−0.870	0.039	29.09	0.03	−3.27	0.04	M20
Pleiades/NGC 2516	…	…	…	−0.870	0.120	29.19	0.28	−3.08	0.27	N16
DT Vir (GJ 494)	2.89	0.03	1T	−0.864	0.039	29.48	0.55	−2.85	0.66	R, X
UCAC4 587-034710	3.05	0.03	M20	−0.854	0.039	29.24	0.09	−3.24	0.11	M22
UCAC4 681-049380	3.22	0.03	M20	−0.844	0.039	29.01	0.04	−3.12	0.05	M20
1RXS J085922.2+403351	3.25	0.03	M20	−0.842	0.039	29.29	0.25	−3.20	0.30	M20
OT Ser (GJ 9520)	3.37	0.03	M20	−0.834	0.039	28.94	0.19	−3.31	0.24	R, X
HAT 169-07364	3.42	0.03	M20	−0.831	0.039	29.56	0.07	−2.91	0.08	M20
PM J23045+4014	3.56	0.04	M20	−0.823	0.039	29.02	0.05	−3.20	0.06	M20
EPIC 206349327	3.72	0.04	M20	−0.813	0.039	29.19	0.04	−3.21	0.05	M20
G 80-21	3.88	0.04	M20	−0.803	0.039	28.90	0.19	−3.19	0.23	M22
G 173-18	4.12	0.04	M20	−0.788	0.039	29.55	0.01	−2.81	0.01	M20
FK Aqr (GJ 867A)	4.23	0.04	M20	−0.781	0.039	29.34	0.30	−3.06	0.36	M20
2MASS J05071824+5307115	4.23	0.04	M20	−0.781	0.039	29.05	0.09	−3.26	0.11	M20
HAT 222-10112	4.77	0.05	M20	−0.747	0.040	29.39	0.06	−2.96	0.07	M20
2MASS J04160235+3325301	4.77	0.05	M20	−0.747	0.040	28.80	0.05	−3.22	0.06	M20
BD-17 6768	4.83	0.05	M20	−0.744	0.040	28.67	0.03	−3.33	0.04	M20
au Mic (GJ 803)	4.85	0.05	M20	−0.743	0.040	29.41	0.07	−3.15	0.12	C, R, S, X
EPIC 202059229	5.01	0.05	M20	−0.733	0.040	29.15	0.37	−3.17	0.44	M22
GSC 02083-01558	5.34	0.05	M20	−0.712	0.040	28.65	0.06	−3.27	0.07	M20
1RXS J175750.3+471634	5.45	0.05	M20	−0.705	0.040	29.09	0.35	−3.36	0.42	M20
LP 149-56	6.17	0.06	M20	−0.661	0.041	28.84	0.03	−3.23	0.04	M20
NGC 6475	…	…	…	−0.660	0.100	29.41	0.33	−2.89	0.36	N16
TYC 3059-299-1	6.46	0.06	M20	−0.643	0.041	29.30	0.02	−3.16	0.02	M20
CD-72 1700A (GJ 1264)	6.67	0.07	M20	−0.629	0.041	29.31	0.17	−2.92	0.20	M22
UCAC4 678-130516	6.85	0.07	M20	−0.618	0.042	29.04	0.11	−3.50	0.13	M20
TYC 2703-706-1	7.80	0.21	M20	−0.559	0.044	29.14	0.06	−3.34	0.07	R, S
UCAC4 666-047939	7.85	0.08	M20	−0.556	0.043	28.88	0.11	−3.49	0.13	M20
EPIC 202059204	7.89	0.08	M20	−0.554	0.043	28.62	0.04	−3.61	0.05	M20
1RXS J051125.0+265849	8.09	0.08	M20	−0.541	0.043	29.00	0.08	−3.08	0.10	M20
UCAC4 729-006249	8.35	0.08	M20	−0.525	0.043	29.09	0.03	−3.33	0.04	M20
TYC 3425-1133-1	8.35	0.08	M20	−0.525	0.043	28.92	0.02	−3.33	0.02	M20
2MASS J21154192+1746242	9.02	0.09	M20	−0.484	0.044	29.06	0.06	−3.31	0.07	M20
UCAC4 643-115616	9.65	0.10	M20	−0.444	0.045	29.25	0.07	−3.34	0.08	M20
CK Ari	9.70	0.10	M20	−0.441	0.045	29.06	0.02	−3.26	0.02	M20
UCAC4 654-053944	10.08	0.28	M20	−0.418	0.048	28.83	0.32	−3.66	0.38	C, R, S, X
UCAC4 669-046011	10.35	0.10	M20	−0.401	0.046	28.70	0.14	−3.60	0.17	M20
TYC 3483-856-1	10.54	0.29	M20	−0.389	0.049	28.80	0.02	−3.47	0.02	R, S
EPIC 210741091	10.95	0.11	M20	−0.364	0.047	28.88	0.02	−3.54	0.02	M20
EPIC 210707811	11.77	0.12	M20	−0.313	0.048	28.79	0.04	−3.34	0.05	R, X
GJ 3367	12.05	0.12	M20	−0.295	0.048	28.79	0.57	−3.65	0.68	M22
V2689 Ori	12.29	0.12	M20	−0.280	0.049	28.45	0.12	−3.97	0.15	M22, R
M37	…	…	…	−0.280	0.050	29.11	0.22	−3.13	0.24	N16
BD+16 2708 (GJ 569 A)	13.68	0.14	M20	−0.194	0.051	28.33	0.43	−3.79	0.52	C, R, X
StKM 1-1215	13.91	0.38	M20	−0.180	0.056	28.48	0.15	−4.03	0.18	R, X
DS Leo (GJ 410)	14.52	0.02	1A	−0.142	0.052	28.35	0.14	−3.99	0.17	M22
Hyaeds/Praesepe	…	…	…	−0.137	0.070	28.76	0.47	−3.58	0.49	N16
1RXS J133332.0+364153	15.36	0.15	M20	−0.090	0.054	28.82	0.22	−3.47	0.26	M20
TYC 3142-1028-1	15.65	0.43	M20	−0.072	0.060	28.67	0.28	−3.35	0.34	R, X
HD 11507	15.80	0.16	M20	−0.063	0.055	27.76	0.25	−4.69	0.30	M22
UCAC4 650-018148	16.14	0.16	M20	−0.041	0.055	28.50	0.18	−4.01	0.22	M20
StKM 1-1407	16.15	0.16	M20	−0.041	0.055	28.70	0.02	−3.76	0.02	M20
GJ 3942	16.30	0.45	GA19	−0.031	0.061	27.81	0.05	−4.53	0.06	F, G
GJ 685	16.30	0.16	M20	−0.031	0.055	27.82	0.30	−4.52	0.36	M20
GJ 338A	16.30	2.40	GA19	−0.031	0.159	27.99	0.40	−4.53	0.48	C, R
G 115-72	16.61	0.17	M20	−0.012	0.056	28.49	0.08	−3.97	0.10	M20

Table 2
(Continued)

Star Name	P_{rot}	Uncert.	P_{rot} src	Log Age	Unc.	$\log(\langle L_X \rangle)$	Unc.	$\log(\langle L_X \rangle / L_{bol})$	Unc.	X-Ray src
GJ 338B	16.61	0.04	GA19	−0.012	0.055	27.96	0.34	−4.47	0.41	C, W
GJ 694.2	17.30	0.17	M20	0.031	0.057	27.98	0.15	−4.36	0.18	M20
EPIC 202059231	17.43	0.17	M20	0.039	0.057	28.46	0.04	−3.66	0.05	M20
GJ 3822	18.30	0.18	M20	0.093	0.059	27.95	0.17	−4.44	0.20	M20
GJ 49	19.45	0.06	1R	0.164	0.060	27.57	0.04	−4.70	0.07	X
GJ 606	20.00	0.20	M20	0.198	0.062	27.86	0.13	−4.29	0.16	M20
1RXS J165257.4+255744	20.22	0.20	M20	0.212	0.063	28.47	0.46	−3.85	0.55	R, X
GJ 2	21.20	0.21	M20	0.273	0.065	27.60	0.13	−4.62	0.16	M20
GJ 720A	34.50	0.35	M20	0.517	0.074	27.71	0.74	−4.71	0.74	M22
GJ 47	34.70	0.35	M20	0.518	0.074	27.42	0.14	−4.45	0.17	M20
HD 38529c	35.90	0.36	M20	0.530	0.076	27.05	0.15	−4.97	0.18	M20
GJ 740	36.40	0.36	M20	0.534	0.077	27.45	0.14	−4.95	0.17	M20
GJ 832	38.10	0.10	1S	0.550	0.080	26.59	0.28	−5.38	0.34	R, S, X
EPIC 201675315	38.13	0.38	M20	0.550	0.081	26.97	0.03	−5.38	0.04	M20
HD 285968 (GJ 176)	40.13	1.13	1R	0.569	0.085	27.25	0.33	−4.88	0.40	C, R
BD-18 359	44.51	0.45	M20	0.610	0.093	27.45	0.67	−4.65	0.67	R, M22
LHS 373 (GJ 552)	44.90	1.20	1R	0.614	0.095	27.50	0.19	−4.68	0.23	M20
GJ 15A	44.95	0.55	1R	0.614	0.095	26.41	0.33	−5.52	0.40	M20
GJ 752A	46.60	0.20	1H	0.630	0.098	26.91	0.17	−5.12	0.21	R
GJ 625	51.22	0.21	1R	0.673	0.108	26.82	0.17	−4.82	0.22	R
BR Psc (GJ 908)	53.70	0.80	1A	0.696	0.114	26.47	0.11	−5.48	0.13	M20
GJ 476	55.00	5.50	SM18	0.708	0.127	27.05	0.15	−4.82	0.22	GA19
HD 95735b (GJ 411)	56.15	0.27	DA19	0.719	0.119	27.07	0.19	−5.05	0.23	M20, X
GJ 1	56.80	5.60	SM17	0.725	0.131	26.71	0.28	−5.27	0.34	R
GJ 625	60.70	0.30	1R	0.761	0.129	26.90	0.14	−4.82	0.17	M20
GJ 433	72.60	0.50	1F	0.873	0.156	26.51	0.31	−5.58	0.37	S, M22

Note. Measured in this study using data from A: APT; F: Feng et al. (2020); M: MEarth; R: RCT; S: Skynet; Z: ZTF; H11: Hartman et al. (2011); SM15: Suárez Mascareño et al. (2015); N16: Núñez & Agüeros (2016); D19: Díaz et al. (2019); GA19: González-Álvarez et al. (2019); M20: Magaudda et al. (2020); and M22: Magaudda et al. (2022). For X-ray sources: C: Chandra; R: ROSAT; S: Swift; and X: XMM.

(This table is available in machine-readable form.)

substantial weakening of their magnetic fields. Eventually, though, rotation slows to the point where it sufficiently impacts the convective turbulence "α effect" mechanism, whereby the magnetic field begins to weaken and X-ray activity begins to decline. However, this is simply a qualitative description of how the dynamo models may explain the data and relationships as presented. A full, theoretical explanation is outside the scope of the current paper.

For a direct measure of UV activity, the Lyα emission line has been used as it represents the majority of all UV flux emitted from cool stars, especially so for M dwarfs. The Lyα database (provided in Tables 4 and 5) consists of far fewer measures compared to the X-ray, though it is no less important as Lyα probes a different layer of the stellar atmosphere and allows accurate stellar wind measures. Obtaining these are crucial for gauging the likelihood that an exoplanet will be able to retain its atmosphere. Figure 4 plots the evolution of Lyα activity over time, using the same early and mid–late M dwarf subsets as for the X-ray relationships. As with the X-ray relationships, the best-fitting parameters for the Lyα data are provided in Table 1. The lack of data is immediately visible in the plots, and the increased uncertainty of the fits. The inflection points are more poorly defined, as is the saturation phase for the early subset where only three measures define this first track. Consequently, this diminishes the reliability of the fit for the second track, as well, though the current fit for this phase is not unreasonable for such a limited data set. For the mid–late M dwarfs, a similar lack of data near the transition between the saturation and declining-activity phases leaves ambiguity in the determination of the inflection point. However, the fit does appear to track the data for the older stars, which makes the current estimate of ∼1.5 Gyr for the duration of the saturation phase a reasonable one. With the limited data currently available, the mid–late subset appears to have a slightly higher level of Lyα activity, relative to bolometric luminosity, when compared to the early subset. This agrees with Linsky et al. (2020), who also reported a trend of increasing activity with decreasing effective temperature.

The final activity index we have analyzed for this study is the $\log R'_{HK}$ index, derived from the Ca II H and K emission features at ∼3968.5 Å and ∼3933.7 Å, respectively. As this measure can be obtained from the ground, there is a greater availability of instruments and observing time. As a result, the Ca II database (provided in Tables 6 and 7) is much more extensive than Lyα, and begins to rival the X-ray database, even though there have not yet been any sensitivity-limited all-sky Ca II surveys for M dwarfs as RÖSAT carried out in the X-ray regime. Due to the large-scale spectroscopic surveys searching for planets around low-mass stars that have come online in recent years, a particular advantage of the Ca II database is that it contains a substantial sample of targets with time-series measures. Many of these surveys have determined or confirmed rotation periods for their M dwarf targets using the repeat measures of activity indices such as Ca II. The

Table 3
Mid–Late M Dwarf X-Ray Data

Star Name	P_{rot}	Unc.	P_{rot} src	Log Age	Unc.	$\log(\langle L_X \rangle)$	Unc.	$\log(\langle L_X \rangle / L_{bol})$	Unc.	X-Ray Src
Pleiades/NGC 2516	…	…	…	−0.870	0.120	28.98	0.31	−2.92	0.35	N16
LP 263-64	2.06	0.02	M20	−0.774	0.019	28.70	0.10	−2.95	0.12	M22
1RXS J175537.7+485740	2.24	0.02	M20	−0.764	0.019	28.57	0.07	−3.08	0.08	M20
BD+20 2465 (AD Leo)	2.24	0.02	1R, M20	−0.764	0.019	28.81	0.05	−3.14	0.06	M22
2MASS J16243140+4549570	2.36	0.02	M20	−0.758	0.019	28.60	0.55	−2.86	0.66	M20
LP 302-37	2.60	0.03	M20	−0.744	0.019	29.26	0.08	−2.98	0.10	M20
1RXS J031028.8+285952	2.87	0.03	M20	−0.729	0.019	28.40	0.13	−3.38	0.16	M20
Ross 154 (GJ 729)	2.87	0.03	M20	−0.729	0.019	27.91	0.20	−3.28	0.24	C, R, X
LP 195-34	3.39	0.03	M20	−0.700	0.019	28.99	0.70	−2.70	0.84	M20, S
1RXS J222653.9+354610	3.55	0.04	M20	−0.691	0.019	28.70	0.10	−2.98	0.12	M20
2MASS J16182491+4437110	4.07	0.04	M20	−0.662	0.019	28.44	0.08	−2.99	0.10	M20
NGC 6475	…	…	…	−0.660	0.100	28.91	0.15	−3.10	0.17	N16
1RXS J174409.9+473444	4.45	0.04	M20	−0.640	0.019	28.46	0.08	−3.26	0.10	M20, S
UCAC4 610-011032	5.85	0.06	M20	−0.562	0.020	29.39	0.02	−2.79	0.02	M20
HAT 214-03527	6.55	0.07	M20	−0.523	0.020	28.50	0.13	−3.34	0.16	M20
2MASS J22170696+3627164	7.04	0.07	M20	−0.495	0.021	28.62	0.08	−3.11	0.10	M20
CW UMa	7.78	0.08	M20	−0.454	0.021	28.61	0.18	−3.04	0.22	M22
BL Lyn	7.85	0.08	1T, M22	−0.450	0.021	28.88	0.25	−3.17	0.30	M22
2MASS J08445708+4640261	8.81	0.09	M20	−0.396	0.022	29.16	0.03	−2.63	0.04	M20
LP 197-37	9.36	0.09	M20	−0.365	0.022	28.35	0.07	−3.50	0.08	M20
G 176-13	10.75	0.11	M20	−0.287	0.023	29.17	0.07	−2.76	0.08	M20
M37	…	…	…	−0.284	0.050	29.11	0.47	−3.14	0.40	N16
HAT 170-05945	10.86	0.11	M20	−0.281	0.023	28.83	0.13	−3.28	0.16	M20
HAT 136-02868	11.99	0.12	M20	−0.217	0.024	28.91	0.08	−3.02	0.10	C, R
HAT 216-04245	12.49	0.12	M20	−0.189	0.025	28.70	0.14	−3.14	0.17	R, X
1RXS J040731.6+340525	12.54	0.13	M20	−0.187	0.025	28.90	0.10	−3.10	0.12	M20
Hyades/Praesepe	…	…	…	−0.155	0.069	28.41	0.49	−3.37	0.53	N16
PM J04310+3647	13.26	0.13	M20	−0.146	0.025	28.37	0.08	−3.45	0.10	M20
G 123-35	13.59	0.14	M20	−0.128	0.026	28.65	0.06	−3.02	0.07	M20
EPIC 210434976	2.55	0.03	M20	−0.097	0.031	28.81	0.46	−2.58	0.55	F, S
YZ CMi (GJ 285)	2.78	0.03	M20	−0.092	0.031	28.50	0.01	−3.15	0.01	M20
EPIC 204957517	2.82	0.03	M20	−0.091	0.031	28.61	0.10	−2.84	0.12	M22
LP 331-57A	14.54	0.15	M20	−0.074	0.027	28.53	0.05	−3.21	0.06	M20
UCAC4 617-130729	14.75	0.15	M20	−0.063	0.027	29.11	0.51	−3.40	0.61	M20
HAT 122-01032	4.38	0.04	M20	−0.052	0.031	28.91	0.01	−2.94	0.01	M20
UCAC3 242-88245	15.10	0.15	M20	−0.043	0.027	28.61	0.09	−3.22	0.11	M22
EPIC 206262336	9.71	0.10	M20	0.082	0.035	28.26	0.03	−2.95	0.04	M20
HG 8-1	17.57	0.18	M20	0.096	0.030	28.20	0.21	−3.61	0.25	M20
EPIC 201482319	11.68	0.12	M20	0.132	0.037	28.50	0.04	−3.19	0.05	M20
G 74-25	15.29	0.15	M20	0.222	0.041	28.47	0.05	−3.30	0.06	R, X
UCAC4 619-055248	19.84	0.20	M20	0.223	0.032	28.41	0.20	−3.39	0.24	F
Ross 868 (GJ 669A)	20.95	0.03	1R	0.285	0.031	28.51	0.29	−3.33	0.35	C, R
UCAC4 692-060564	21.52	0.22	M20	0.317	0.034	28.63	0.08	−3.38	0.10	M20
GJ 4334	23.54	0.24	N16	0.449	0.027	28.46	0.24	−3.01	0.29	R, X
CD-40 5404B (GJ 358)	25.26	0.70	1S	0.472	0.031	28.05	0.15	−3.90	0.18	M20
LP 167-71	27.82	0.28	M20	0.483	0.032	28.88	0.03	−3.07	0.04	M20
2MASS J00240376+2626299	29.84	0.30	M20	0.491	0.034	27.87	0.13	−3.53	0.16	M20
1RXS J075235.5+410055	30.33	0.30	M20	0.493	0.034	28.33	0.13	−3.40	0.16	M20
1RXS J174158.2+475403	32.23	0.32	M20	0.500	0.036	28.35	0.12	−3.31	0.14	M20
GJ 674	33.29	0.33	AD17	0.505	0.037	27.52	0.17	−4.26	0.20	R, S, X
2MASS J06170531+8353354	40.75	0.10	1M	0.535	0.043	27.38	0.27	−4.29	0.32	W, R
2MASS J20125995+0112584	41.15	1.23	13	0.537	0.044	27.95	0.49	−3.29	0.59	C, R
GJ 752A	46.60	0.20	1H	0.559	0.049	26.91	0.17	−5.12	0.21	R
GJ 588	51.70	0.10	1H	0.579	0.055	26.92	0.19	−4.67	0.24	R
2MASS J06575703+6219197	54.50	0.55	M20	0.591	0.057	27.83	0.07	−3.56	0.08	M20
2MASS J16492028+5101177	54.71	0.55	M20	0.591	0.058	26.89	0.43	−4.77	0.52	M20
GJ 163	56.04	2.47	1M	0.597	0.060	26.87	0.15	−5.09	0.18	Fo22
2MASS J05011802+2237015	70.67	0.71	M20	0.656	0.076	27.40	0.12	−3.59	0.14	M20
2MASS J19310458-0306186	70.92	0.71	M20	0.657	0.076	26.82	0.11	−3.94	0.13	M20
GJ 436	71.40	0.40	1R	0.659	0.076	26.85	0.46	−5.11	0.55	C, R, X
2MASS J13505181+3644168	72.18	0.72	M20	0.662	0.077	26.84	0.30	−4.62	0.36	M20
2MASS J17311725+8205198	79.22	0.79	M20	0.691	0.085	27.40	0.09	−3.70	0.11	M20
GJ 3253	81.55	0.82	M20	0.700	0.088	26.94	0.04	−4.16	0.05	M20
2MASS J03103891+2540535	82.83	0.83	M20	0.706	0.089	26.95	0.30	−4.45	0.36	M20

Table 3
(Continued)

Star Name	P_{rot}	Unc.	P_{rot} src	Log Age	Unc.	$\log(\langle L_X \rangle)$	Unc.	$\log(\langle L_X \rangle / L_{bol})$	Unc.	X-Ray Src
GJ 357	86.10	0.90	1H	0.719	0.093	26.94	0.25	−4.70	0.30	X
G 184-31	86.25	0.86	M20	0.719	0.093	26.72	0.09	−4.41	0.11	M20
EPIC 206019392 (G 876)	87.95	0.61	1A	0.726	0.095	26.34	0.10	−5.38	0.12	F, X
GJ 551	88.98	0.01	1S	0.730	0.096	26.52	0.39	−4.27	0.47	W
2MASS J18551471-7115026	90.69	0.91	M20	0.737	0.098	26.63	0.30	−4.95	0.36	M20
2MASS J14211512-0107199	91.41	0.91	M20	0.740	0.099	26.27	0.30	−5.13	0.36	M20
2MASS J05015746-0656459	92.15	2.76	M20	0.743	0.101	27.32	0.08	−3.82	0.10	C, R
2MASS J19194119-5955194	92.69	0.93	M20	0.745	0.100	26.36	0.30	−5.05	0.36	M20
2MASS J23351050-0223214	100.40	1.00	M20	0.777	0.110	26.05	0.43	−4.71	0.52	M20
GJ 667C	103.10	0.60	1H	0.788	0.112	26.69	0.28	−5.06	0.34	C, R
2MASS J20403364+1529572	104.60	1.05	M20	0.794	0.114	26.60	0.13	−4.62	0.16	M20
GJ 628	108.70	1.50	1R	0.810	0.119	26.40	0.55	−5.21	0.66	C, R
EPIC 201518346	112.83	1.13	M20	0.827	0.124	26.74	0.03	−4.42	0.04	M20
EPIC 205913009 (GJ 1265)	116.40	1.16	M20	0.841	0.129	26.03	0.03	−5.11	0.04	M20
2MASS J11505787+4822395	117.50	1.18	M20	0.846	0.129	26.20	0.30	−4.92	0.36	M20
GJ 447	124.95	1.90	1R	0.876	0.139	26.71	0.36	−4.88	0.43	M22
2MASS J19204795-4533283	133.90	1.34	M20	0.912	0.149	26.51	0.30	−4.66	0.36	M20
GJ 699	149.50	0.60	1AR,3	0.968	0.069	25.53	0.54	−5.57	0.65	F20, R
GJ 273	160.83	2.48	1R	0.968	0.069	26.36	0.32	−5.23	0.38	R
GJ 581	147.80	0.60	1AR	0.968	0.069	25.73	0.55	−5.94	0.66	C, S
GJ 191	153.20	3.70	1S	1.021	0.046	26.46	0.50	−5.23	0.60	C, R, S, X

Note. Measured in this study using data from A: APT; F: Feng et al. (2020); M: MEarth; R: RCT; S: Skynet; and Z: ZTF. Ages calculated using the M4+ young track relationship. H11: Hartman et al. (2011); N16: Núñez & Agüeros (2016); PL16: Loyd et al. (2016); AD17: Astudillo-Defru et al. (2017); W18: Wright et al. (2018); N18: Newton et al. (2018); D19: Díaz et al. (2019); T19: Toledo-Padrón et al. (2019); M20: Magaudda et al. (2020); M22: Magaudda et al. (2022); F20: France et al. (2020); F22: Freund et al. (2022); and Fo22: Foster et al. (2022). C: Chandra; R: ROSAT; S: Swift; and X: XMM.

(This table is available in machine-readable form.)

Table 4
Early M Dwarf Lyα Data

Star Name	P_{rot} (days)	Unc.	Src	Log Age (Gyr)	Unc.	$\log(L_{Ly\alpha}/L_{bol})$	Unc.
au Mic	4.86	0.05	K19	−0.742	0.040	−3.500	0.019
HIP 23309	8.60	0.07	M10	−0.510	0.043	−4.163	0.087
GJ 410	14.52	0.02	1R	−0.142	0.052	−3.306	0.922
GJ 49	19.45	0.06	1R	0.164	0.060	−3.931	0.804
GJ 649	21.42	0.20	1R	0.286	0.065	−4.044	0.300
GJ 3470	21.54	0.49	Ko19	0.294	0.071	−4.170	0.300
GJ 887	33.00	0.91	AD17	0.503	0.072	−4.077	0.300
GJ 205	33.72	0.13	1R	0.509	0.072	−4.112	0.300
GJ 832	38.10	0.11	1S	0.550	0.081	−4.635	0.032
GJ 176	39.30	0.10	1AR	0.561	0.083	−4.515	0.026
GJ 15A	50.69	0.21	1R	0.668	0.107	−4.302	0.300
GJ 411	56.15	0.27	D19	0.719	0.119	−4.834	0.300

Note. Measured in this study using data from A: APT; M: MEarth; R: RCT; S: Skynet; Z: ZTF; K19: Küker et al. (2019); Ko19: Kosiarek et al. (2019); DA19: Díez Alonso et al. (2019); AD17: Wargelin et al. (2017); and M10: Messina et al. (2010).

(This table is available in machine-readable form.)

evolution of M dwarf Ca II activity over time is plotted in Figure 5 (again, using the same M dwarf subsets as for the X-ray and Lyα relationships) and the best-fitting paramaeters are provided in Table 1.

As Figure 5 shows, the richer data sets allow for a better understanding of the levels of activity during the saturation phase. According to the fits, early M dwarfs appear slightly more active during both their saturation phase and at very old ages. Both subsets show a ∼1.5 order of magnitude decrease in Ca II activity; less than the decrease they experience in X-ray activity. This phenomenon of chromospheric activity experiencing a less drastic decline with age has also been observed in other studies (Linsky et al. 2020; Loyd et al. 2021; Pineda et al. 2021). One potential theory put forth is that cool star chromospheres are acoustically heated, and the coronae are magnetically heated. This hints at a potentially interesting

Table 5
Mid–Late M Dwarf Lyα Data

Star Name	P_{rot} (days)	Unc.	src	log Age (Gyr)	Unc.	log ($L_{Ly\alpha}/L_{bol}$)	Unc.
AD Leo (GJ 388)	2.24	0.02	1R, Mo08	−0.764	0.019	−3.576	0.013
HIP 112312	2.355	0.005	M10	−0.758	0.019	−3.170	0.075
YZ CMi (GJ 285)	2.7758	0.0002	Mo08	−0.092	0.031	−3.218	0.300
GJ 729	2.90	0.03	KS12	−0.727	0.019	−3.483	0.300
HIP 17695	3.87	0.01	M10	−0.673	0.019	−3.356	0.064
EV Lac (GJ 873)	4.3715	0.0002	Mo08	−0.052	0.031	−3.769	0.095
GJ 4334	23.54	0.24	N16, DA19	0.449	0.027	−3.828	0.882
LHS 2686	28.80	0.80	N16, DA19	0.486	0.034	−4.020	0.300
GJ 674	33.30	1.00	KS07	0.505	0.037	−4.060	0.300
GJ 163	56.04	2.47	1M	0.597	0.060	−4.326	0.300
GJ 588	61.30	6.50	AM15	0.618	0.070	−4.153	0.300
GJ 436	71.40	0.40	1R	0.659	0.076	−4.590	0.070
GJ 849	71.95	1.01	1R	0.661	0.077	−4.432	0.300
GJ 876	87.30	5.70	1A	0.723	0.097	−4.678	0.051
Proxima Cen (GJ 551)	88.98	0.01	1S	0.730	0.096	−3.795	0.008
GJ 667C	103.10	0.60	1H	0.788	0.112	−4.227	0.083
GJ 1214	125.00	5.00	Ma18	0.876	0.140	−4.592	0.297
GJ 1132	129.15	3.87	N18	0.893	0.144	−4.535	0.249
GJ 581	132.50	6.30	SM15	0.968	0.069	−4.957	0.126
Barnard's Star (GJ 699)	149.50	0.60	1R	0.968	0.069	−4.746	0.009
Luyten's Star (GJ 273)	160.83	2.48	1R	0.968	0.069	−4.213	0.300
Kapteyn's Star (GJ 191)	153.20	3.70	1S	1.061	0.046	−4.969	0.300

Note. Measured in this study using data from A: APT; H: HARPS-Trifonov; M: MEarth; R: RCT; S: Skynet; Z: ZTF; Mo08: Morin et al. (2008); M10: Messina et al. (2010); KS13: Kiraga & Stępień (2013); SM15: Suárez Mascareño et al. (2015); N16: Newton et al. (2016); Ma18: Mallonn et al. (2018); N18: Newton et al. (2018); and DA19: Díez Alonso et al. (2019) .

(This table is available in machine-readable form.)

scenario where the chromospheres and coronae of early and mid–late M dwarfs are, when compared to each other, declining by the same relative amounts even though (a) one subset is partially convective and the other is fully convective and (b) the chromospheres are acoustically heated (driven by convection), yet the coronae of the two subsets should be heated by different convection/rotation contributions. In studying several far-ultraviolet emission features, Loyd et al. (2021) further found that transition region activity levels may behave intermediate to those of the chromosphere and corona, perhaps indicating how the different heating contributions vary between the atmospheric layers. Linsky et al. (2020) found that FGKM stars all display similar levels of saturated coronal activity, but as they age they display an intensifying trend of increasing coronal activity with decreasing effective temperature. As mentioned previously, Linsky et al. (2020) also reported a trend of increasing Lyα activity with decreasing effective temperature, but the trend was essentially consistent across stellar age and not as steep as the coronal activity trend for their oldest target group. Further, Linsky et al. (2020) proposed additional theories, including that flare heating could play a more prominent role in the atmospheres of lower-mass stars, or that the higher surface gravities of M dwarfs could lead to different atmospheric structures, higher photospheric gas pressure, and, in turn, stronger magnetic fields.

The chromospheric activity of early M dwarfs exits the saturation phase after ~600 Myr, where coronal activity exits after ~500 Myr. This difference is not conclusively determined and the chromospheric and coronal activity levels can be considered to exit the saturation phase at a similar age to each other. The mid–late M dwarf inflection point is not strictly defined by the fits, since the Ca II database also suffers from a (albeit less substantial) lack of data near this age range. For this subset, the Ca II data show that chromospheric activity levels exit the saturation phase at ~1.4 Gyr, where coronal activity levels take ~2.3 Gyr to exit this phase. This is a more substantial difference, but again one that is owed heavily to the lack of near-inflection Ca II data. Further data at this important age range could reduce, possibly even eliminate, the current disparity between the lengths of the chromospheric and coronal saturation phases. Firmly establishing the chromospheric and coronal inflection points can help shed light on the outer atmospheric heating mechanisms at work in M dwarfs. It is encouraging to see that the chromospheric saturation phase lengths derived from Ca II data are close to those obtained from the much more sparse Lyα data.

To aid planetary atmosphere and habitability studies better, we have converted our X-ray ($\log(L_X/L_{bol})$) relationships into cumulative X–UV (~5–1700 Å) irradiance relationships. To do this, SEDs constructed by the MUSCLES and Mega-MUSCLES surveys (France et al. 2016; Loyd et al. 2016; Youngblood et al. 2016; Froning et al. 2019; Wilson et al. 2021) were obtained from MAST[2] (France 2016) and used to determine integrated fluxes over the 5–1700 Å range. A linear relationship was determined to be,

$$\log(L_{X-UV(5-1700\text{Å})}/L_{bol}) = 0.5728[0.0589] \times \log(L_X/L_{bol}) - 1.0509[0.2921]. \quad (2)$$

Using this equation, we were able to analyze the X–UV irradiances of M dwarfs over time and determine the segmented

[2] https://archive.stsci.edu/prepds/muscles/

Table 6
Early M Dwarf Ca II Data

Star Name	P_{rot}	Unc.	P_{rot} src	log Age	Unc.	log $\langle R_{HK} \rangle$	Unc.
GJ103	1.56	0.02	AD17	−0.947	0.038	−3.877	0.100
Pleiades	…	…	−0.870	0.120	−4.170	0.290	…
GJ 494	2.89	0.03	AD17	−0.864	0.039	−3.930	0.130
GJ 9520	3.37	0.03	AD17	−0.834	0.039	−3.978	0.750
GJ 867A	4.23	0.04	AD17	−0.781	0.039	−4.149	0.120
GJ 182	4.41	0.04	AD17	−0.770	0.040	−3.865	0.450
GJ 803	4.85	0.05	AD17	−0.743	0.040	−4.022	0.210
GJ 1264	6.67	0.07	AD17	−0.629	0.041	−3.994	0.190
GJ 208	11.49	0.36	M22	−0.330	0.052	−4.140	0.625
GJ 3685	11.60	0.12	P23	−0.323	0.048	−4.229	0.402
GJ 3367	12.05	0.12	AD17	−0.295	0.048	−4.088	1.157
DS Leo(GJ 410)	14.52	0.02	1A	−0.142	0.052	−4.123	0.352
Hyades	…	…	…	−0.137	0.070	−4.510	0.360
GJ 3942	16.3	0.1	SM18	−0.031	0.055	−4.550	0.100
GJ 338A	16.3	2.4	SM18	−0.031	0.159	−4.455	0.100
GJ 338B	16.61	0.04	R23	−0.012	0.055	−4.470	0.100
GJ 373	17.18	1	1T	0.023	0.084	−4.423	0.057
GJ 694.2	17.3	0.1	SM18	0.031	0.057	−4.460	0.150
GJ 21	17.4	1.1	SM18	0.037	0.089	−4.720	0.030
GJ 3822	18.3	0.1	SM18	0.093	0.058	−4.700	0.040
GJ 49	19.45	0.06	1R	0.164	0.060	−4.401	0.588
GJ 606	20	2	SM18	0.198	0.138	−4.890	0.060
GJ 3218	21.10	0.21	R23	0.267	0.065	−4.517	0.276
GJ 2	21.2	0.5	GA19	0.273	0.071	−4.622	0.540
GJ 382	21.20	0.10	SM16	0.273	0.064	−4.655	0.247
GJ 649	21.34	0.29	1R	0.282	0.066	−4.621	0.695
GJ 3470	21.54	0.49	K19	0.294	0.071	−4.883	0.210
GJ 685	21.83	3.76	1A	0.312	0.242	−4.596	0.036
GJ 450	22.8	1	DA19	0.372	0.091	−4.622	0.392
GJ 9404	23.2	0.1	SM18	0.403	0.049	−4.710	0.040
GJ 846	26.30	5.60	SM15	0.440	0.080	−4.820	0.080
GJ 4057	26.7	0.1	SM18	0.444	0.061	−5.030	0.040
GJ 4306	27	2.5	SM18	0.447	0.066	−4.840	0.040
GJ 229	27.3	0.1	SM16	0.450	0.062	−4.773	0.275
GJ 740	27.51	0.07	1AS	0.451	0.062	−4.578	0.367
GJ 514	30	0.9	SM17	0.475	0.067	−4.879	0.127
GJ 1030	32	3	SM18	0.493	0.075	−4.840	0.060
GJ 162	32.4	1.6	SM18	0.497	0.071	−4.950	0.040
GJ 3998	33.6	3.6	SM18	0.508	0.079	−5.010	0.050
GJ 205	33.72	0.13	1R	0.509	0.072	−4.743	0.393
GJ 393	34.15	0.34	AD17	0.513	0.074	−5.031	0.106
GJ 720A	34.5	4.7	SM18	0.517	0.086	−4.859	0.170
GJ 47	34.7	0.1	SM18	0.518	0.074	−4.910	0.060
GJ 9689	35.7	0.2	SM18	0.528	0.076	−4.820	0.050
GJ 548A	36.6	0.1	SM18	0.536	0.078	−4.780	0.030
GJ 3997	37	13	SM18	0.540	0.144	−4.780	0.030
GJ 880	37.5	0.38	AD17	0.544	0.079	−4.741	0.585
GJ 832	38.1	0.11	1S	0.550	0.079	−5.148	0.172
GJ 2066	40.7	0.41	AD17	0.574	0.086	−5.220	0.077
GJ 676A	41.2	3.8	SM15	0.579	0.094	−5.033	0.110
GJ 156.1A	41.2	6.3	SM18	0.579	0.105	−5.040	0.040
GJ 3138	42	0.42	AD17	0.587	0.088	−4.855	0.126
GJ 552	43.5	1.5	SM18	0.601	0.092	−5.070	0.040
GJ 15A	44.99	3.4	1R	0.615	0.099	−5.270	0.100
GJ 184	45	0.1	SM18	0.615	0.095	−5.090	0.040
GJ 412A	46.4	0.2	SM18	0.628	0.097	−5.460	0.150
GJ 9440	48	4.8	SM18	0.643	0.111	−5.250	0.040
GJ 654	49	0.49	AD17	0.652	0.103	−5.266	0.093
GJ 521A	49.5	3.5	SM18	0.657	0.109	−5.360	0.050
GJ 536	49.8	0.1	1H	0.660	0.104	−5.059	0.130
V*BR Psc (GJ 908)	49.9	3.5	SM18	0.661	0.110	−5.380	0.060
GJ 119A	51.2	4.4	SM18	0.673	0.115	−4.970	0.040
GJ 625	51.22	0.21	1R	0.673	0.108	−5.333	1.273
GJ 526	52.3	0.4	1H	0.683	0.110	−5.113	0.089

Table 6
(Continued)

Star Name	P_{rot}	Unc.	P_{rot} src	log Age	Unc.	log $\langle R_{HK} \rangle$	Unc.
GJ 476	55	5.5	SM18	0.708	0.127	−5.210	0.050
GJ 411	56.15	0.27	DA19	0.719	0.119	−5.371	0.100
GJ 1	56.8	5.6	SM17	0.725	0.131	−5.431	0.121
GJ 433	72.6	0.5	1F	0.873	0.156	−5.289	0.384

Note. Measured in this study using data from A: APT; F: Feng et al. (2020); H: Trifonov et al. (2020); M: MEarth; R: RCT; S: Skynet; Z: ZTF; KS13: Kiraga & Stępień (2013); SM15: Suárez Mascareño et al. (2015); SM16: Suárez Mascareño et al. (2016); N16: Newton et al. (2016); SM17: Suárez Mascareño et al. (2017); AD17: Astudillo-Defru et al. (2017); N18: Newton et al. (2018); SM18: Suárez Mascareño et al. (2018); DA19: Díez Alonso et al. (2019); and B22: Boudreaux et al. (2022).

Each star's log $\langle R_{HK} \rangle$ value was determined using the combined data (depending on availability) from Pace (2013), Suárez Mascareño et al. (2015, 2016), Astudillo-Defru et al. (2017), Suárez Mascareño et al. (2017), Boro Saikia et al. (2018), Suárez Mascareño et al. (2018), Melbourne et al. (2020), Perdelwitz et al. (2021), Boudreaux et al. (2022), Ibañez Bustos et al. (2023), and Marvin et al. (2023). The Pleiades and Hyades values were obtained from Fang et al. (2018).

(This table is available in machine-readable form.)

Table 7
Mid–Late M Dwarf Ca II Data

Star Name	P_{rot}	Unc.	P_{rot} src	log Age (Gyr)	Unc.	log $\langle R_{HK} \rangle$	Unc.
Pleiades	…	…	…	−0.870	0.120	−4.201	0.260
LP 831-35	2.08	0.02	N18	−0.773	0.019	−4.234	0.002
AD Leo (GJ 388)	2.24	0.02	1R	−0.764	0.019	−4.191	0.070
LP 805-1	2.81	0.03	N18	−0.732	0.019	−4.390	0.004
GJ 729	2.87	0.03	AD17	−0.729	0.019	−4.405	0.510
LP 709-18	3.15	0.03	N18	−0.713	0.019	−4.540	0.002
UCAC3 151-377752	6.72	0.07	N18	−0.513	0.021	−4.491	0.002
EPIC 201482319	11.68	0.12	M20	−0.235	0.024	−4.228	0.402
Hyades	…	…	…	−0.137	0.070	−4.330	0.480
LP 735-29	2.10	0.02	N18	−0.109	0.031	−4.405	0.003
G 70-46	2.35	0.02	N16	−0.103	0.031	−4.473	0.002
GJ 285	2.78	0.03	AD17	−0.092	0.031	−4.151	0.510
LP 851-399	3.07	0.03	N18	−0.084	0.031	−4.295	0.003
GJ 3127	14.79	0.15	N18	−0.060	0.027	−4.791	0.003
G 152-1	4.38	0.04	N18	−0.052	0.031	−4.621	0.002
GJ 4274	4.41	0.30	1T	−0.051	0.032	−4.350	0.250
LHS 1358	4.73	0.05	N16	−0.043	0.031	−4.635	0.001
SCR J1107-3420B	7.61	0.08	N18	0.030	0.033	−4.225	0.003
BL Lyn	16.40	0.16	R23	0.030	0.028	−4.280	0.049
GJ 4071	8.16	0.25	1T	0.043	0.034	−4.292	0.221
G 151-64	8.43	0.08	N18	0.050	0.034	−4.202	0.002
GJ 669A	20.95	0.03	1R	0.285	0.031	−4.570	0.400
GJ 479	23.75	0.24	AD17	0.454	0.028	−4.766	0.284
GJ 362	24.87	0.04	1R	0.471	0.031	−4.493	0.436
LP 740-10	25.01	0.25	N18	0.471	0.031	−4.764	0.003
GJ 358	25.26	0.70	1S	0.472	0.031	−4.655	0.077
LHS 5273	30.40	0.30	N18	0.493	0.034	−4.609	0.004
GJ 674	33.29	0.33	AD17	0.505	0.037	−4.978	0.210
GJ 832	34.78	2.82	1S	0.511	0.040	−5.148	0.172
GJ 176	40.14	0.57	1R	0.533	0.043	−4.949	0.091
GJ 3293	41.00	0.40	AD17	0.536	0.044	−5.114	0.070
LP 734-34	42.99	0.43	N18	0.544	0.045	−4.689	0.003
GJ 752A	46.60	0.20	1H	0.559	0.049	−5.113	0.102
GJ 889B	51.20	0.51	N18	0.577	0.054	−5.278	0.008
GJ 588	51.70	0.52	1H	0.579	0.055	−5.143	0.403
GJ 877	52.80	0.53	SM16	0.584	0.056	−5.180	0.293
GJ 1088	53.74	0.54	N18	0.587	0.057	−5.251	0.014
GJ 163	56.04	2.47	1M	0.597	0.060	−5.422	0.130
GJ 618A	56.52	0.60	AD17	0.599	0.060	−5.401	0.070
WT 244	66.45	0.66	N18	0.639	0.071	−5.603	0.015
LP 816-60	67.60	0.68	SM16	0.644	0.072	−5.116	0.488
GJ 849	71.05	0.50	AD17	0.657	0.076	−5.054	0.540
GJ 436	71.40	0.40	1R	0.659	0.076	−5.269	0.599
GJ 1148	71.50	5.10	DA19	0.659	0.079	−5.336	1.030
L 154-205	73.14	0.73	N18	0.666	0.079	−4.874	0.005

Table 7
(Continued)

Star Name	P_{rot}	Unc.	P_{rot} src	log Age (Gyr)	Unc.	log $\langle R_{HK} \rangle$	Unc.
GJ 12	78.50	0.80	AD17	0.688	0.084	−5.368	0.070
LHS 610	78.80	0.80	AD17	0.689	0.085	−5.375	0.070
LP 780-32	79.15	0.79	N18	0.691	0.085	−5.252	0.007
L 50-78	80.97	0.81	N18	0.698	0.087	−5.388	0.007
GJ 4132	84.99	0.85	N18	0.714	0.092	−5.653	0.008
GJ 357	86.10	0.90	1H	0.719	0.093	−5.430	0.710
GJ 876	87.95	0.61	1A	0.726	0.095	−5.258	0.260
GJ 3323	88.50	0.89	KS12	0.728	0.095	−4.747	0.436
GJ 551	88.98	0.01	1S	0.730	0.096	−5.134	0.243
GJ 876	91.00	0.90	AD17	0.738	0.099	−5.496	0.070
GJ 3843	91.43	0.91	N18	0.740	0.099	−5.585	0.013
L 127-124	91.62	0.92	N18	0.741	0.100	−5.629	0.010
UCAC4 195-119117	93.70	0.94	N18	0.749	0.102	−5.006	0.008
GJ 836	94.25	0.94	N18	0.752	0.102	−6.195	0.019
GJ 3382	95.00	0.95	B22	0.755	0.103	−5.698	0.019
LP 728-71	99.66	1.00	N18	0.774	0.109	−5.632	0.011
GJ 1057	102.00	1.00	AD17	0.783	0.112	−5.522	0.070
GJ 667C	103.10	0.60	1H	0.788	0.112	−5.500	0.172
GJ 628	108.70	1.50	1R	0.810	0.119	−5.287	0.613
GJ 3963	122.66	1.23	N18	0.867	0.136	−5.596	0.011
GJ 447	124.95	1.90	1R	0.876	0.139	−5.235	0.505
PM J08446-4805	129.51	1.30	N18	0.894	0.143	−5.668	0.007
L 210-11	136.92	1.37	N18	0.924	0.153	−5.998	0.018
GJ 581	147.80	0.60	1A+R	0.968	0.069	−5.487	0.425
GJ 699	149.50	0.60	1A+R	0.968	0.069	−5.490	0.634
GJ 273	160.83	2.48	1R	0.968	0.069	−5.273	0.520
LP 855-60	154.89	1.55	N18	0.997	0.174	−5.963	0.025
GJ 191	153.2	3.7	1S	1.061	0.046	−5.609	0.495

Note. Measured in this study using data from A: APT; F: Feng et al. (2020); H: Trifonov et al. (2020); M: MEarth; R: RCT; S: Skynet; Z: ZTF; KS13: Kiraga & Stępień (2013); SM15: Suárez Mascareño et al. (2015); SM16: Suárez Mascareño et al. (2016); N16: Newton et al. (2016); SM17: Suárez Mascareño et al. (2017); AD17: Astudillo-Defru et al. (2017); N18: Newton et al. (2018); SM18: Suárez Mascareño et al. (2018); DA19: Díez Alonso et al. (2019); and B22: Boudreaux et al. (2022).

Each star's log $\langle R_{HK} \rangle$ value was determined using the combined data (depending on availability) from Pace (2013), Suárez Mascareño et al. (2015, 2016), Astudillo-Defru et al. (2017), Suárez Mascareño et al. (2017), Boro Saikia et al. (2018), Suárez Mascareño et al. (2018), Melbourne et al. (2020), Perdelwitz et al. (2021), Boudreaux et al. (2022), Ibañez Bustos et al. (2023), and Marvin et al. (2023). The Pleiades and Hyades values were obtained from Fang et al. (2018).

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fit parameters shown in Table 1. These relationships have not been plotted, as they mirror those of the $\log(L_X/L_{bol})$ activity over time (save for the relative shifting/scaling of values). At present, since X-ray data were used as the basis for scaling, the saturation phases again last for ∼500 Myr and ∼2.3 Gyr for the early and mid–late M dwarfs, respectively.

5. Conclusions

The *Living with a Red Dwarf* (*LivRed*) program has constructed activity–age relationships for M dwarfs using three different tracers of magnetic activity, along with a broad X–UV band (5–1700 Å). Multiple measures of each star's activity levels were gathered whenever possible to help mitigate scatter due to stellar variability, and many of the ages were calculated using the age–rotation relationships presented in Engle & Guinan (2023) and shown in Figure 1.

The stars were divided into two subsets, as previously observed (Guinan & Engle 2018; Mullan & Houdebine 2020) and utilized when constructing the age–rotation relationships—what we have called the "early" [M0–2] and "mid–late" [M2.5–6.5] subsets. Both subsets show similar drops in relative activity levels as they evolve. The most significant difference between the two subsets' activity levels is the duration of their initial saturation phases. Early M dwarfs experience a coronal (X-ray) and chromospheric (Ca II) saturation phase of ∼500–600 Myr. Mid–late dwarfs, by contrast, experience a coronal saturation phase of ∼2.3 Gyr and a chromospheric saturation phase that is currently estimated to last ∼1.4–1.5 Gyr, but is not as well constrained by either the Ca II or Lyα data. As proposed by other studies in the literature, the disparity in coronal versus chromospheric activity behaviors for mid–late M dwarfs could potentially indicate that the atmospheric plasmas are being heated by different mechanisms, and further study along these lines is encouraged.

For both subsets, however, the extended initial periods of enhanced X–UV activity will be important to account for when analyzing the potential habitability of any planets orbiting M dwarfs. For example, the publicly available `VPlanet`[3] software has a number of useful routines dealing with exoplanet habitability calculations (do Amaral et al. 2022), but the default prescription for high-energy stellar evolution is

[3] https://github.com/VirtualPlanetaryLaboratory/vplanet

that of Ribas et al. (2005) which was constructed for solar-type G dwarfs and uses a saturation phase length of 100 Myr. This can significantly underestimate the saturation phase of M dwarfs, and the data and relationships presented here will help future studies to account for such differences better.

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Software: astropy (Astropy Collaboration et al. 2013, 2018, 2022), Matplotlib (Hunter 2007), Pandas (Wes McKinney 2010), NumPy (Harris et al. 2020), and SciPy (Virtanen et al. 2020).

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